1. Create an SQL view named `DoctorAppointmentView` that stores information about doctors and their appointments. The view should include the doctor's full name, specialization, and the total number of appointments they have handled.

   CREATE VIEW DoctorAppointmentView AS
   SELECT CONCAT(d.FirstName, ' ', d.LastName) AS DoctorFullName,
   d.Specialization, COUNT(a.AppointmentID) AS TotalAppointments
   FROM Doctors d
   JOIN Appointments a ON d.DoctorID = a.DoctorID
   GROUP BY d.DoctorID, d.FirstName, d.LastName, d.Specialization;

2. Retrieve all staff members whose first and last names end with the same letter.

   SELECT StaffID, CONCAT(FirstName, ' ', LastName) AS FullName, Role,
   Department FROM Staff
   WHERE RIGHT(FirstName, 1) = RIGHT(LastName, 1);

3. Retrieve a list of patients along with the doctors who treated them **sorted by recent appointments first**. The query should include the patient's first name, **patient's age** and only the first **3 characters of the doctor's first name** in **upper case.**

   SELECT   p.FirstName, datediff(YY,p.DOB,GETDATE()) AS PatientAge,
   UPPER(LEFT(d.LastName,3)) AS DocName
   FROM Appointments a
   JOIN Patients p ON a.PatientID = p.PatientID
   JOIN Doctors d ON a.DoctorID = d.DoctorID
   ORDER BY a.AppointmentDate DESC;

4. Write a simple sql query to get patient's information who are still admitted in the hospital (not discharged).

   SELECT * FROM Patients p
   JOIN Admissions a ON p.PatientID = a.PatientID
   WHERE a.DischargeDate IS NULL;

5. Retrieve all staff members whose first name matches the first name of another staff member with the same role. (sub-query)

```
SELECT StaffID, FirstName, LastName, Role
FROM Staff s1
WHERE FirstName IN (
   SELECT FirstName
   FROM Staff s2
   WHERE s1.Role = s2.Role AND s1.StaffID<>s2.StaffID
);
```

Results | Messages

	StaffID	FirstName	LastName	Role
1	S001	John	Clark	Nurse
2	S007	John	Wick	Nurse

6. Retrieve a list of patients whose first name contains the second last letter 'V' and consists of at least 3 characters. Also update the doctor name (doctors table) to replace DR. with the Doctor.

```
SELECT PatientID, CONCAT(FirstName, ' ', LastName) AS FullName
FROM Patients
WHERE FirstName like '%_V_'

UPDATE Doctors
SET FirstName = REPLACE(FirstName, 'Dr.', 'Doctor')
WHERE FirstName LIKE 'Dr.%';
```

7. Retrieve patients who have been admitted more than once to either 'Private' or 'ICU' rooms, along with their total days hospitalized. Only include completed admissions (where discharge dates are recorded). Show the patient's full name, number of admissions, and total days hospitalized.

```
SELECT CONCAT(p.FirstName, ' ', p.LastName) AS PatientFullName,
   COUNT(a.AdmissionID) AS NumberOfAdmissions,
   SUM(DATEDIFF(DAY, a.AdmissionDate, a.DischargeDate)) AS
TotalDaysHospitalized
FROM Patients p
JOIN  Admissions a ON p.PatientID = a.PatientID
JOIN  Rooms r ON a.RoomID = r.RoomID
WHERE r.RoomType IN ('Private', 'ICU') AND
        a.DischargeDate IS NOT NULL  -- Only completed admissions
GROUP BY  p.PatientID, p.FirstName, p.LastName
```

```
HAVING COUNT(a.AdmissionID) > 1;
```

8. Provide the no. of appointments for **all** the patients in the database.

```
SELECT CONCAT(p.FirstName, ' ', p.LastName) AS PatientFullName,
   COUNT(a.AppointmentID) AS NumberOfAppointments
FROM Patients p LEFT JOIN Appointments a ON p.PatientID = a.PatientID
GROUP BY p.PatientID, p.FirstName, p.LastName;
```

9. Provide a list of patients whose date of birth is earlier than all other patients in the database who were born in the same year.

```
SELECT p1.PatientID, CONCAT(p1.FirstName, ' ', p1.LastName) AS
PatientFullName, p1.DOB, YEAR(p1.DOB) AS BirthYear
FROM Patients p1
WHERE p1.DOB < ALL ( SELECT p2.DOB FROM Patients p2 WHERE
YEAR(p1.DOB) = YEAR(p2.DOB) AND p1.PatientID <> p2.PatientID )
```

10. Create a new table called CurrentInpatients to track patients who are currently hospitalized. The table should include:
    i) InpatientID (Integer, Auto-increment)
    ii) PatientId (Integer, Foreign key)
    iii) PatientName (String, constraint: not null)
    iv) Gender (String, default value: ‘unknown’, should only accept three values (male, female, unknown))
    v) DOB (string of size 50 characters)
    Composite Primary key: InpatientID and PatientId
    After creating the table, populate it with data for existing patients who are currently hospitalized such that Birth date is saved in format ‘month day, year’ like March 25, 2025 (without using joins).

```
-- Step 1: Create the table with all constraints
CREATE TABLE CurrentInpatients (
   InpatientID INT identity(1,1),
   PatientID INT NOT NULL,
   PatientName VARCHAR(100) NOT NULL,
   Gender VARCHAR(10) DEFAULT 'unknown'
      CHECK (Gender IN ('male', 'female', 'unknown')),
   DOB VARCHAR(50),
```

```
    CONSTRAINT PK_CurrentInpatients PRIMARY KEY (InpatientID, PatientID),
    CONSTRAINT FK_CurrentInpatients_Patients FOREIGN KEY (PatientID)
        REFERENCES Patients(PatientID)
);

-- Step 2: Insert data for currently hospitalized patients with formatted DOB
INSERT INTO CurrentInpatients ( PatientID, PatientName, Gender, DOB)
SELECT p.PatientID, CONCAT(p.FirstName, ' ', p.LastName) , p.Gender,
    FORMAT(p.DOB, 'MMMM dd, yyyy')
FROM  Patients p
WHERE p.PatientID IN ( SELECT a.PatientID
FROM Admissions a
WHERE a.DischargeDate IS NULL);  -- Currently hospitalized
```